

Linux & Red Hat Enterprise Notes for Exam Preparation

I. What is Linux?

- **Definition:** An open-source, Unix-like operating system kernel.
- **Core:** The heart of the Linux operating system.
- **Open Source:** Its source code is freely available, modifiable, and distributable.
- **Kernel:** It's often referred to as "Linux," but Linux specifically refers to the kernel itself. The full operating system is usually called GNU/Linux.

II. Why Learn Linux?

- **Versatility:** Used in servers, supercomputers, embedded systems, mobile devices (Android), and desktops.
- **Stability & Reliability:** Known for its robustness and uptime.
- **Security:** Generally considered more secure than other operating systems due to its permission model and active community.
- **Cost-Effective:** Free to use and distribute; reduces software licensing costs.
- **Flexibility & Customization:** Highly customizable to specific needs.
- **Developer-Friendly:** Strong tools and environments for programming.
- **Career Opportunities:** High demand for Linux administrators and developers in the IT industry.

III. Where is Linux Used?

- **Servers:** Dominates the server market (web servers, database servers, application servers).
- **Cloud Computing:** Backbone of major cloud platforms (AWS, Azure, GCP).
- **Supercomputers:** Powers almost all of the world's supercomputers.
- **Embedded Systems:** Used in routers, smart TVs, IoT devices, car infotainment systems.
- **Mobile Devices:** Android is based on the Linux kernel.
- **Desktops:** Available as various distributions (e.g., Ubuntu, Fedora).
- **Scientific Research:** Common in academic and research environments.

IV. History of Linux

- **1991:** Linus Torvalds, a Finnish student, created the initial Linux kernel.
- **Inspiration:** Based on Minix, a Unix-like operating system.
- **GNU Project:** Linux kernel combined with GNU utilities to form a complete OS.
- **Growth:** Evolved through community contributions and commercial ventures (e.g., Red Hat).

V. Distros (Distributions)

- **Definition:** A complete operating system built on top of the Linux kernel, bundled with system utilities, libraries, and application software.
- **Examples:**
 - **Ubuntu:** User-friendly, popular for desktops and servers.
 - **Fedora:** Community-driven, upstream of Red Hat, focused on latest technologies.
 - **Debian:** Stable, foundational for many other distros (including Ubuntu).
 - **Red Hat Enterprise Linux (RHEL):** Commercial, stable, widely used in enterprises, comes with paid support.
 - **CentOS:** (Historically) Free, community-supported rebuild of RHEL source code. Now replaced by CentOS Stream.
 - **Kali Linux:** Designed for penetration testing and digital forensics.
 - **Slackware:** One of the oldest Linux distributions, known for simplicity and stability.
 - **SUSE:** Commercial Linux distribution, especially SUSE Linux Enterprise Server (SLES).
 - **Linux Mint:** Popular, user-friendly desktop distribution based on Ubuntu/Debian.

VI. Kernel

- **Definition:** The core component of the operating system.
- **Function:** Manages system resources, hardware interaction, and process scheduling.
- **Role:** Acts as a bridge between hardware and software.
- **Types:** Monolithic (Linux is primarily monolithic, but with modularity).

VII. What is Shell?

- **Definition:** A command-line interpreter that provides an interface for users to interact with the operating system.
- **Types:** Bash (Bourne Again SHell - most common), Zsh, Csh, Ksh.
- **Shell Prompt:**
 - `[root@host ~]#` (for root user, `~` indicates home directory, `#` indicates root user)
 - `[user@host ~]$` (for regular user, `$` indicates regular user)
- **Syntax of Command:**
 - `command <options> argument /etc.`
 - **Command:** The program to be executed (e.g., `ls`, `cd`, `mkdir`).
 - **Options/Flags:** Modify the behavior of the command (e.g., `-l` for long listing in `ls`). Typically start with `-` or `--`.
 - **Argument:** The input the command acts upon (e.g., a file name, a directory path).

VIII. Basic Commands (Examples)

- `ls` : List directory contents.
 - `ls -l` : Long listing format.
 - `ls -a` : List all files, including hidden ones.
- `cd` : Change directory.
 - `cd ..` : Move up one directory.
 - `cd ~` : Go to home directory.
- `pwd` : Print working directory (show current directory path).
- `mkdir` : Make directory.
 - `mkdir new_folder`
- `rmdir` : Remove empty directory.
- `touch` : Create empty file or update timestamp.
- `cp` : Copy files or directories.
 - `cp file1 file2`
 - `cp -r dir1 dir2` (for directories)
- `mv` : Move or rename files/directories.
- `rm` : Remove files or directories.
 - `rm file.txt`
 - `rm -r folder/` (recursively remove folder and its contents)
 - `rm -rf folder/` (force recursive removal, use with caution!)
- `cat` : Concatenate and display file content.
- `more` / `less` : View file content page by page.
- `head` / `tail` : Display beginning/end of a file.
- `grep` : Search for patterns in text.
- `man` : Display the manual page for a command.
- `sudo` : Execute a command as a superuser (root).
- `ps` : Display information about running processes.
- `top` : Display dynamic real-time view of running processes.
- `ping` : Test network connectivity.
- `ip addr` (or `ifconfig`) : Display network interface information.
- `ssh` : Secure Shell for remote login.